

TURNING POINT

SMOKELESS POWDER

In 1884, the French chemist Paul Vieille invented a new propellant—“smokeless powder.” Unlike gunpowder—the propellant used universally up to this point—smokeless powder did not obscure the battlefield or give away a concealed shooter’s position. Being smokeless also meant that it left little residue to clog the barrels and actions of guns. Also, crucially, it burned more slowly and generated greater power. These advantageous properties combined to have a profound effect on the development of firearms. A key step was the creation of the first machine-gun—the Maxim gun (see pp.184–85).



5.56MM-CALIBER CARTRIDGE

▲ SMOKELESS POWDER

All modern cartridges, such as this 5.56mm NATO, contain smokeless powder as a propellant. Smokeless powder is composed of a mixture of nitrocellulose and other chemicals. It is shaped into thin flakes before being loaded into the cartridges.

Gunpowder, or black powder, was a mix of saltpeter, sulfur, and charcoal. It produced thick white smoke on burning, obscured targets, and clogged up the barrels and mechanisms of guns. Highly combustible, it could explode when unconfined, leading to accidents. These problems were overcome with Vieille’s smokeless powder, with the added bonus of more power.

USING SMOKELESS POWDER

The French government was the first to take advantage of the remarkable ballistic properties of smokeless powder, developing the *Le fusil de 8mm Modèle 1886*—the Lebel rifle—named after the designer of its cartridge, Colonel

“... as they used smokeless powder, it was almost impossible to see them...”

THEODORE ROOSEVELT, ON THE SPANISH IN THE SPANISH-AMERICAN WAR (1898)

Nicholas Lebel of France. This true modern rifle used Lebel’s 8mm cartridge with a lead bullet encased in a jacket of cupro-nickel or copper, containing the smokeless propellant. It was faster and weighed less than its predecessors. The cartridge had a flat nose so that it would be safe nose-to-tail in the tubular magazine (see p.116) of the Lebel rifle.

NEW WEAPONS

In conjunction with the metallic cartridge (see pp.112–13), smokeless powder spurred the development of powerful firearms, notably machine-guns such as the Maxim gun (see pp.184–85), and new forms of artillery with greatly improved performance. It left little residue, which allowed the bore and workings of guns to be built to a perfect fit, making weapons such as infantry rifles more accurate. There was also less risk of a bullet jamming in the bore, which would be disastrous with a gun firing several rounds per second. Smokeless powder also provided more propulsive force than the same amount of gunpowder, which significantly increased the effective range of weapons as faster projectiles had a flatter trajectory. It burned clean with little smoke, giving shooters a clear field of vision and allowed them to fire shots with a fair amount of accuracy while hidden from view.

By the turn of the century, bullet designs had begun to be refined to exploit the properties of this new propellant. Captain Desaleux’s solid brass pointed (spitzer) bullet used smokeless

powder and had a tapering “boat-tail,” which increased its velocity, giving it a flatter trajectory and improving its long-range performance. It was the first bullet of its type to be placed into service by any army and it heralded the development of modern bullets.

Smokeless powder was seen in action in the battles in and around Colenso (1899–1900) on the Tugela River in the second Anglo-Boer War.

» BEFORE

Gunpowder burned fast, coating the bores and actions of guns with a thick layer of “fouling.” Also, when exposed to moisture in the air, this “fouling” corroded the insides of the barrels.



GUNPOWDER

● DIFFICULTY IN PINPOINTING ENEMIES

on the battlefield through billowing smoke made it difficult to gauge tactics and plan countermeasures.

● ACCUMULATION OF FOULING, or residue, in a gun’s barrel would make the gun increasingly inaccurate and reduce its range. Severe fouling could jam the gun’s action, or cause a bullet to get jammed in its bore.

● FURTHER DEVELOPMENT of firearms was impeded by limitations in gunpowder’s ballistic and chemical properties.

KEY FIGURE

Paul Marie Eugène Vieille
(1854–1934)

Paul Vieille was a chemistry graduate of Ecole Polytechnique. He became director of the “Laboratoire Central des Poudres et Salpêtres” in Paris as well as a member of the French Academy of Sciences. In recognition of his invention of smokeless powder, he was awarded the Leconte prize of 50,000 Francs by the French Academy of Sciences in 1889.

